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Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

Urban traffic congestion causes delays, fuel wastage, and increased air pollution. Traditional traffic monitoring systems often rely on centralized data collection, which increases network communication and raises data privacy concerns.

Key Issues:

- Real-time traffic data collection at multiple junctions is challenging.
- Centralized processing can increase network usage and latency.
- Sharing raw traffic data raises privacy concerns.
- Traffic prediction models require continuous updates to adapt to changing conditions.
- Frequent federated learning updates can increase communication costs.
- Traffic signals often lack predictive, data-driven optimization.

Background

SmartFlow AI combines Internet of Things (IoT), Artificial Intelligence (AI), and Federated Learning to improve urban traffic management. ESP32 devices connected to IR and ultrasonic sensors collect vehicle data at individual junctions, enabling real-time traffic monitoring.

- **IoT-Based Monitoring:** Sensors measure vehicle counts and traffic conditions at each junction.
- **Local AI Prediction:** Edge devices or connected edge computers predict congestion using traffic data, speed, weather, and time.
- **Federated Learning:** Each junction trains a local model and shares model updates with a central server instead of raw traffic data.
- **Communication Cost Evaluation:** Upload/download size, latency, bandwidth usage, and communication rounds are measured.
- **Smart Traffic Control:** Predicted congestion can support adaptive traffic signal timing.

Literature & RFC Review

1. Federated Averaging (FedAvg), 2017
McMahan et al. introduced FedAvg, enabling multiple devices to train models locally and share model updates instead of raw data.

2. Federated Learning in Intelligent Transportation Systems, 2025
Research highlights federated learning for traffic-flow prediction while identifying privacy, bandwidth limitations, and differences in local traffic data as key challenges.

3. Communication-Efficient Traffic Prediction, 2024
Research on FedCE explores gradient compression and adaptive aggregation to reduce communication overhead in federated traffic prediction.

Proposed Solution

SmartFlow AI is an IoT-based traffic management system that combines real-time sensor data, local AI prediction, and federated learning to reduce congestion and communication overhead.

System Workflow:

- **IoT Data Collection:** ESP32 devices connected to IR and ultrasonic sensors collect vehicle counts and traffic measurements at each junction.
- **Data Transmission:** Sensor readings are published locally using MQTT for monitoring and processing.
- **Local AI Prediction:** Each edge node predicts traffic conditions for the next 15 minutes using vehicle count, speed, weather, and time.
- **Federated Learning:** Local models train on junction-specific data and send model weights to a central server without sharing raw traffic data.
- **Global Model Aggregation:** The server combines model updates using Federated Averaging (FedAvg) and distributes the improved model to participating nodes.

Proposed Solution

- **Communication Cost Evaluation:** Upload/download size, latency, bandwidth usage, and communication rounds are measured.
- **Smart Dashboard:** A web dashboard displays traffic conditions, congestion predictions, model performance, and communication costs.
- **Adaptive Signal Control:** Predicted congestion guides traffic signal timing adjustments to improve traffic flow.

Optimization Proposed

SmartFlow AI focuses on reducing communication overhead while maintaining accurate traffic predictions and responsive signal control.

Key Optimizations:

Reduced Communication Rounds: Perform multiple local training steps before sending model updates.
Model Compression: Compress model weights to reduce upload and download data.
Selective Client Participation: Prioritize junctions with significant traffic changes or useful model updates.
Adaptive Update Frequency: Adjust synchronization frequency according to traffic conditions and model performance.
Edge-Based Prediction: Generate local predictions to reduce dependence on continuous cloud communication.

Optimization Proposed

Smart Signal Timing: Adjust green-light duration based on predicted congestion, subject to safe timing limits.

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.

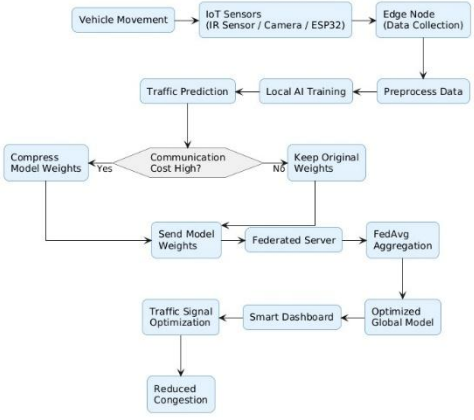


Figure: System Architecture UML Diagram

Data Flow Diagram

Overview of the architecture. Include the flow chart diagram created earlier. Visual: Diagrams or architecture visuals.

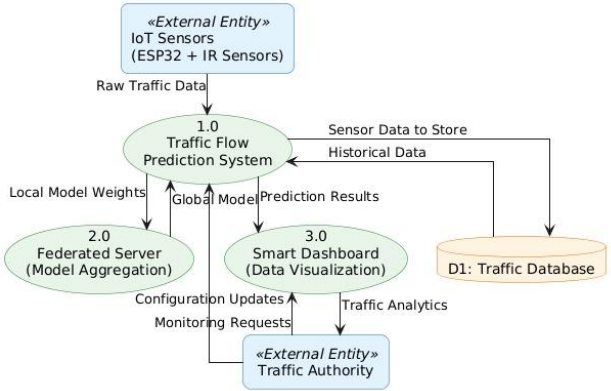


Figure: Data Flow Diagram (DFD)

Deployment Timeline

SmartFlow AI will be developed in phases, progressing from IoT data collection to federated learning and intelligent traffic control.

Phase 1: IoT Setup
Connect ESP32 devices with IR and ultrasonic sensors to collect vehicle data.
Phase 2: Data Collection & MQTT
Transmit sensor readings using MQTT and prepare traffic datasets.
Phase 3: Local AI Prediction
Develop models to predict traffic congestion at individual junctions.
Phase 4: Federated Learning
Implement local model training and global aggregation using Flower and FedAvg.
Phase 5: Communication Optimization
Apply model compression and reduced communication rounds; measure bandwidth and latency.

Deployment Timeline

Phase 6: Dashboard & Signal Control
Build a dashboard for traffic visualization and demonstrate adaptive signal timing.
Phase 7: Testing & Evaluation
Compare prediction accuracy, communication costs, and traffic-flow performance.
Final Deliverable: A working SmartFlow AI prototype demonstrating IoT-based traffic prediction, federated learning, communication-cost evaluation, and adaptive traffic signal control.

Bibliography I

1. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, “Communication-Efficient Learning of Deep Networks from Decentralized Data,” *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, vol. 54, pp. 1273–1282, 2017. Available: https://proceedings.mlr.press/v54/mcmahan17a.html

2. “A Survey on Federated Learning in Intelligent Transportation Systems,” *IEEE Transactions on Intelligent Vehicles*, vol. 10, no. 5, pp. 3043–3059, 2025. DOI: 10.1109/TIV.2024.3446319.

3. F. Gao, C. Zhang, J. Qiao, K. Li, and Y. Cao, “Communication-Efficient Wireless Traffic Prediction with Federated Learning,” *Mathematics*, vol. 12, no. 16, article 2539, 2024. DOI: 10.3390/math12162539.

Thank You for Your Attention!